

☹ You CANNOT use this or any other formula sheet on the Assessment. You need to memorize this stuff

Algorithms

Here's a list of some functions listed from fast to slow:

$\mathcal{O}(1)$ **Constant**

$\mathcal{O}(\log(n))$ **Logarithmic.**

Ignore Log bases: $\log_{10}(n) = \log(n)$

Same growth as $\log(n^k) = k \log(n)$

$\mathcal{O}(\log^c(2n))$ **Polylogarithmic**

Don't forget that: $\log^c x = (\log x)^c$.

$\mathcal{O}(n)$ **Linear**

$\mathcal{O}(n \log(2n))$ **Linearithmic/Loglinear**

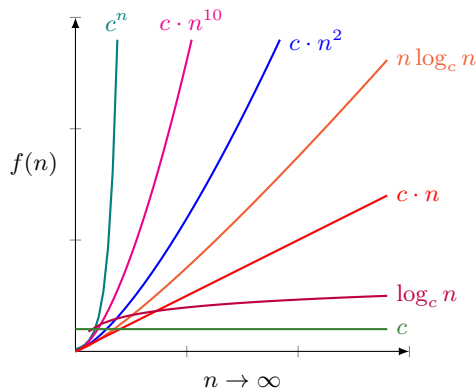
$\mathcal{O}(n^2)$ **Quadratic**

$\mathcal{O}(n^2 \log(2n))$

$\mathcal{O}(n^3)$ **Polynomial**

$\mathcal{O}(2^n)$ **Exponential**

$\mathcal{O}(n!)$



Some properties

We ignore coefficients and log bases, but not exponent bases.

ex. $3 \cdot 5^x$ is $\mathcal{O}(5^x)$

When adding functions, we only take the slowest function's Big- $\mathcal{O}$.

ex. $0.01x^3 + 12x^2 - 2 \log 37x$ is $\mathcal{O}(n^3)$

When multiplying functions, we take the product of the functions' Big- $\mathcal{O}$.

ex. $x^2 + 2 \in \mathcal{O}(n^2)$ and $\log_{10}(x) \in \mathcal{O}(\log n)$ so then $(x^2 + 2)(\log_{10}(x))$ is $\mathcal{O}(n^2 \log(n))$

Advanced Counting

Combination Number ways to choose k objects from a group of n when *order doesn't matter*:

$${}_nC_k = \binom{n}{k} = \frac{n!}{n-k!k!}$$

Permutation Number ways to choose k objects from a group of n when *order does matter*:

$${}_nP_k = \frac{n!}{n-k!}$$

Binomial Theorem

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

This gives us the following identity:

$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots (-1)^n \binom{n}{n} = 0$$

Discrete Probability

- $P(A)$ is the probability of event A happening.
- $P(\bar{A}) = 1 - P(A)$ is the probability of event A **not** happening.
- $P(A|B)$ is the probability of event A *given* that B has already happened. When A and B are independent, $P(A|B) = P(A)$.
- $P(A \cap B) = P(A|B)P(B)$ is the probability of event A *and* B happening.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ is the probability of the A *or* B happening.

Bayes' Theorem: For events X and F from a common sample space where $P(X), P(F) \neq 0$,

$$P(F|X) = \frac{P(X|F)P(F)}{P(X)} \quad (\text{version 1})$$

and equivalently,

$$P(F|X) = \frac{P(X|F)P(F)}{P(X|F)P(F) + P(X|\bar{F})P(\bar{F})} \quad (\text{version 2})$$

While the first version does not appear in our textbook or pre-assessment practice problems, this equivalent definition is listed in the lesson summary.

Note that $P(F \cap X) = P(X|F)P(F)$. So version 1 is often shown as:

$$P(F|X) = \frac{P(X \cap F)}{P(X)} \quad (\text{version 3})$$

Expected Value The *expected value* of a discrete random variable X is

$$E[X] = \sum_{s \in S} X(s)p(s)$$

where S is the set of possible outcomes of X , $p(s)$ is the probability of outcome s , and $X(s)$ is the value of X for s .

Number Theory

Stuff to add?

- modular arithmetic basics
- needed modular arithmetic properties
- Euclidean algorithm
- Extended Euclidean Algorithm
- Encryption

Recursion and Induction

Stuff to add?

- Label for Induction parts
- Summation formulation label?